LAB NO: 3

Objectives

To learn to use MATLAB and the Symbolic Math Toolbox to (1) find Laplace transforms for time functions, (2) find time functions from Laplace transforms, (3) create LTI transfer functions from symbolic transfer functions, and (4) perform solutions of symbolic simultaneous equations.

Practice Questions

Students who are performing the MATLAB exercises and want to explore the added capability of MATLAB's Symbolic Math Toolbox should now run ch2apF4 in Appendix F at www.wiley.com/go/Nise/ControlSystemsEngineering8e, where Example 2.10 is solved. You will learn how to use the Symbolic Math Toolbox to solve simultaneous equations using Cramer's rule. Specifically, the Symbolic Math Toolbox will be used to solve for the transfer function in Eq. (2.82) using Eq. (2.80).

- Use the following MATLAB and Symbolic Math Toolbox statements to help you solve for the electrical currents in Eq. (2.94).

```
syms s I1 I2 I3 V
A=[(2*s + 2) -(2*s + 1)...
-1 -(2*s + 1) (9*s + 1)...
-4*s -1 -4*s...(4*s + 1 + 1/s)];
B=[I1;I2;I3];
C=[V;0;0];
B=inv(A)*C;
pretty(B)
```

Practical Problem 1:

1. Complete the prelab experiment 2.1, Write a LabVIEW program that executes an equivalent of the following C-like code, where x is an input and y is an output (Formula Nodes are not allowed)

```
if(abs(x) < 0.1)
        y = 1;
else
        if(x >= 0)
             y = 0;
        else
             y = 2;
```

Run your program for the following inputs: x = 0.05, -0.05, 1, -1.

2. Write a LabVIEW program that receives three colors representing a resistor's value and returns the numeric resistor value in ohms. Your interface should be similar to the one shown in Figure 2.1 The third band should include silver and gold colors.

   Run your program at least for the following inputs:

   a. Red Red Black
   b. Brown Black Orange
   c. Orange White Gold

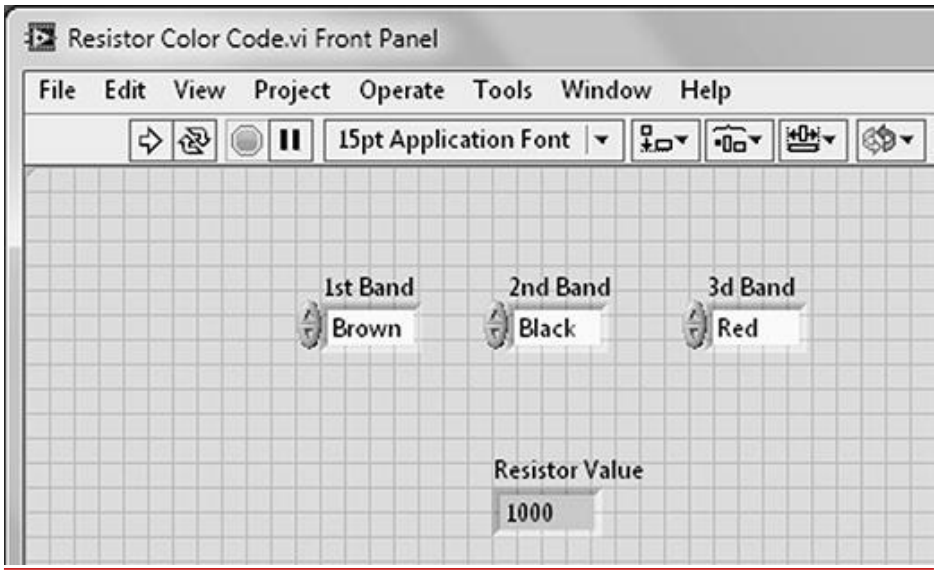


Figure 2.1: LabVIEW VI for resistor color code.

Practical Problem 2:

Complete the Prelab problem 2.2.

1. It is well known that $1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x}$ when $|x| < 1$.

   Write a LabVIEW program that takes as an input a value for x, and a number of iterations. The program will use a loop to calculate the sum of the geometric series for the specified number of operations. It will also calculate the closed-form expression for the series. The program will display the two results, and will also display the absolute error of the difference.

   Demonstrate your program with $x = 0.5$ and 3, 10, and 200 iterations.

2. Write a LabVIEW program that generates a 50% duty-cycle square-wave signal between 0 and X_{max} volts, where $X_{max} < 10V$, with a nonzero variable frequency. The amplitude and frequency will be inputs. Display the waveform on a waveform chart, which will be an output. In this part you are not allowed to use the LabVIEW-provided function generation blocks.

   Demonstrate this program for amplitudes of 1, 5, and 10 V and for 1 Hz and 5 Hz.

Practical Problem 3:

Complete Prelab problem 2.3. Use Figure 2.17.1 in Prelab 2.3 subsection 3, and do the following.

Write a MATLAB script to find symbolic expressions for the currents and transfer function using MATLAB's symbolic toolbox. Ex: In the symbolic toolbox, the current I_2 can be defined as a symbol using `syms I2`

The MATLAB script must do the following:

1) Create the symbolic matrix $\mathbf{A}$. Matrix $\mathbf{A}$ must be stored in the variable `A`

- The elements of `A` must be symbolic and found by writing the mesh equations in matrix form $\mathbf{Ab} = \mathbf{c}$ using the same equation order as in the problem description above.
- MATLAB's `syms s` command defines a symbolic value `s`

2) Calculate the symbolic expression for current $I_1(s)$. The expression must be stored in the variable `I1`

3) Calculate the symbolic expression for current $I_2(s)$. The expression must be stored in the variable `I2`

4) Calculate the symbolic expression for current $I_3(s)$. The expression must be stored in the variable `I3`

5) Calculate the symbolic transfer function $G(s) = \frac{V_o(s)}{V(s)}$. The expression must be stored in the variable `G`